MILITARY SPECIFICATION

SLING, BOMB HOISTING, AERO 61B

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers a group of components which can be assembled to provide slings capable of lifting weapons of varying diameters and weights. Combinations of components permit handling of weapons up to 5,000 pounds and 30 inches in diameter.

1.2 Classification - The Sling shall consist of the components listed in 6.4.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.3 of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Parts and materials - Unless otherwise specified herein, all parts and materials shall conform to the detailed requirements indicated on the applicable drawings.

3.3 Construction - The sling shall be constructed in accordance with drawings listed on IL65A101. No departure from applicable drawings shall be permitted without prior written approval by the procuring activity.

3.4 Interchangeability - All parts having the same part number shall be completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be governed by the drawing requirements of Specification MIL-D-70327. Parts joined together by welding or cast as one piece shall be interchangeable as units or assemblies. Proper gaging methods and facilities shall be maintained to insure that this requirement is fulfilled.

3.5 Performance - Each sling assembly shall be capable of withstanding the proof loading as specified in Section 4, without any malfunction or deformation.

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3.6 General operating requirements - The Bomb Hoisting Sling shall function properly under all tests specified in Section 4.

3.7 Identification of product - The equipment shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship - The workmanship shall be that required by commercial practice, covering this type of equipment, for the quantity production of analogous parts. Particular attention shall be given to freedom from defects, burrs and sharp edges.

3.9 Manufacturing and finish - Manufacturing processes and finishes shall be as shown on the applicable drawings unless otherwise approved by the procuring activity.

3.10 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - The contractor is responsible for the performance of all inspection requirements as specified.

4.1.1 Classification tests - Slings covered by this specification shall be subjected to the following tests to determine compliance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.1.2 Test samples - The slings used for preproduction and sampling tests shall not be considered as equipments under the contract. They are to be supplied as equipments over and above the quantity specified for delivery in the contract. Upon completion of tests, salvageable components will be returned to the contractor for his use in fabrication and/or testing of equipment to be submitted for acceptance.

4.1.3 Test conditions - All tests shall be conducted at room ambient conditions. All load applying equipment shall be calibrated for accuracy within plus or minus 25 pounds. Loads may be applied hydraulically, mechanically, or by deadweight.

4.2 Preproduction tests - Preproduction tests are those tests which are conducted by the government on the first two slings produced from production tools. The contractor shall deliver two preproduction samples to the government laboratory specified in the contractor or order (see 6.5). When the contractor delivers the sample units, a certification from the cognizant Government Inspector that the material used in the samples are in accordance

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with the requirements of this specification shall be forwarded with the samples. Any defects or failures occurring during the government tests shall be cause for tentative rejection of the preproduction samples. All rejected samples may be returned to the contractor for rework or correction of defects when considered practical and feasible by the procuring activity, otherwise the contractor will be required to submit additional samples for retest. Preproduction tests shall include all tests specified below and such other tests as deemed necessary by the procuring activity to determine that equipment meets all the requirements of this specification and the contract:

(1) Examination of product
(2) Operational test
(3) Proof test
(4) Band tensile test.

4.2.1 Preproduction approval - Approval of the preproduction samples shall be by the procuring activity upon satisfactory completion of all tests. No production equipments or lot samples shall be delivered prior to approval of the preproduction samples. Fabrication of production equipment prior to the approval of the preproduction samples is at the contractor's own risk.

4.3 Acceptance tests - Acceptance of equipment shall be on a lot basis. Each lot shall be composed of equipment which has successfully passed the individual tests. Acceptance tests shall consist of the following:

(1) Individual tests
(2) Sampling tests

4.3.1 Individual tests - Individual tests shall be accomplished, under the responsibility of the contractor, on each sling submitted for acceptance under the contract. The contractor shall maintain test reports for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor and this data shall be made available to the procuring activity upon request. The cognizant Government Inspector shall supervise all tests. Individual tests shall consist of the following tests:

(1) Examination of product
(2) Operational test
(3) Proof test.

4.3.2 Sampling tests - Sampling tests shall be conducted at a government laboratory under the responsibility of the procuring activity. Sampling tests shall be conducted on samples selected from lots of equipment which have successfully passed the individual tests. Samples to be tested shall be selected by the Government Inspector in accordance with sampling test schedule in 4.3.2.1. A lot size shall not exceed 500 slings except with the approval of the procuring activity. Samples shall be suitably labeled to identify them with a particular lot. Samples shall be forwarded to the government laboratory specified in the contract or order (Sec. 6.5). Sampling tests shall consist of all tests specified in 4.2.

4.3.2.1 Sampling test schedule - Unless otherwise specified in con-

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tract, the sampling test schedule shall be as follows:

Quantity offered for Lot Acceptance	Quantity to be Tested
99 or less	1
100 - 299	2
300 - 500	3

4.3.2.2 Lot acceptance -- The acceptance of each lot shall be by the procuring activity after satisfactory completion of sampling tests.

4.4 Test methods

4.4.1 Examination of product - Each sling shall be examined carefully to determine that the material, and workmanship requirements have been met.

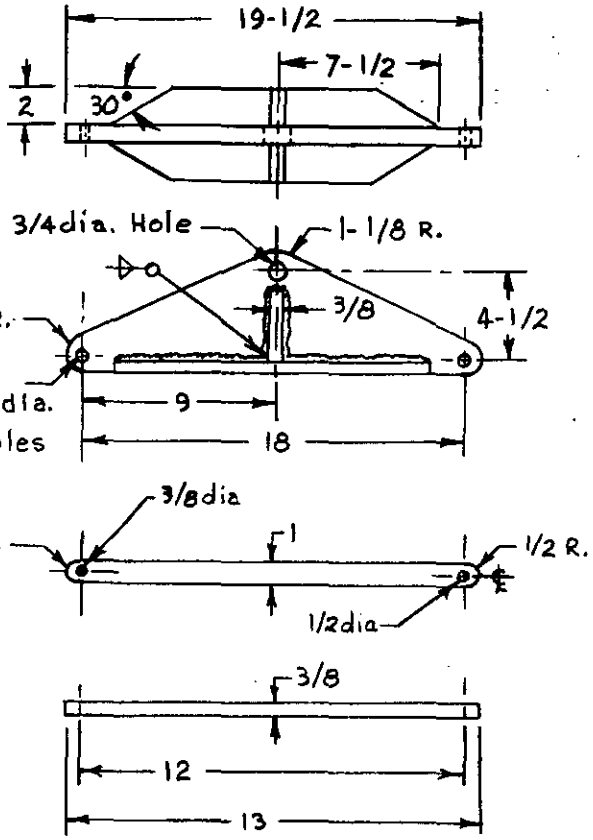
4.4.2 Operational test - In order to determine compatibility of interrelated components, each sling assembly, utilizing in turn the 0.035 inch thick bands and then the 0.045 inch thick bands, shall be installed on a test fixture as shown in Figure 1. All units shall have no misalignment or restrictive conditions in respect to adjustment, affixing of related components to each other, or resistance to latching operation and safety lock function.

4.4.3 Proof test - Each sling shall be assembled using the 0.035 inch thick bands and proof tested by applying a load of 8,000 pounds in the manner shown in Figure 1. After the load has been removed, all parts shall function freely and bands shall be visually inspected to insure that there is no evidence of cracking. Proof testing of the 0.045 inch thick bands shall not be required. Figure 2 shows the test fixture details.

4.4.4 Band tensile test - This test is to be conducted after all other specified tests have been completed. From each band a test specimen, as shown in Figure 3, shall be evaluated for tensile strength and ductility. The minimum acceptable breaking strength of the 0.035 inch and 0.045 inch thick bands shall be 10,400 pounds and 13,000 pounds, respectively. Deformation shall be developed in dart shaped zones located normal to the applied stresses, on opposite sides of each hole between the loading fixtures. Measurements of thickness shall be made at each of points A, B, and C as indicated on Figure 3, at each intermediate hole in the test speciment after testing. The reduction of thickness at point A or B from the thickness at point C shall be not less than 0.001 inch .

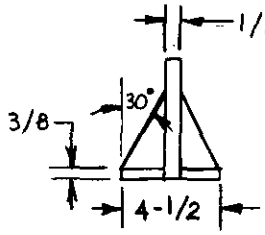
4.5 Rejection and retest - Any unit failing to pass the tests specified in 4.3.1 shall be rejected. Units which have been rejected may be reworked or have parts replaced to correct defects and resubmitted for testing.

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(1) YOKE
Material: Alloy Steel MIL-S-18729

Quantity: One required.



(2) LINK
Material: Alloy Steel MIL-S-18729 Heat treated 180,000psi

Quantity: Two required.

(3) DUMMY STORE
Material: Mild Steel
Quantity: One required.

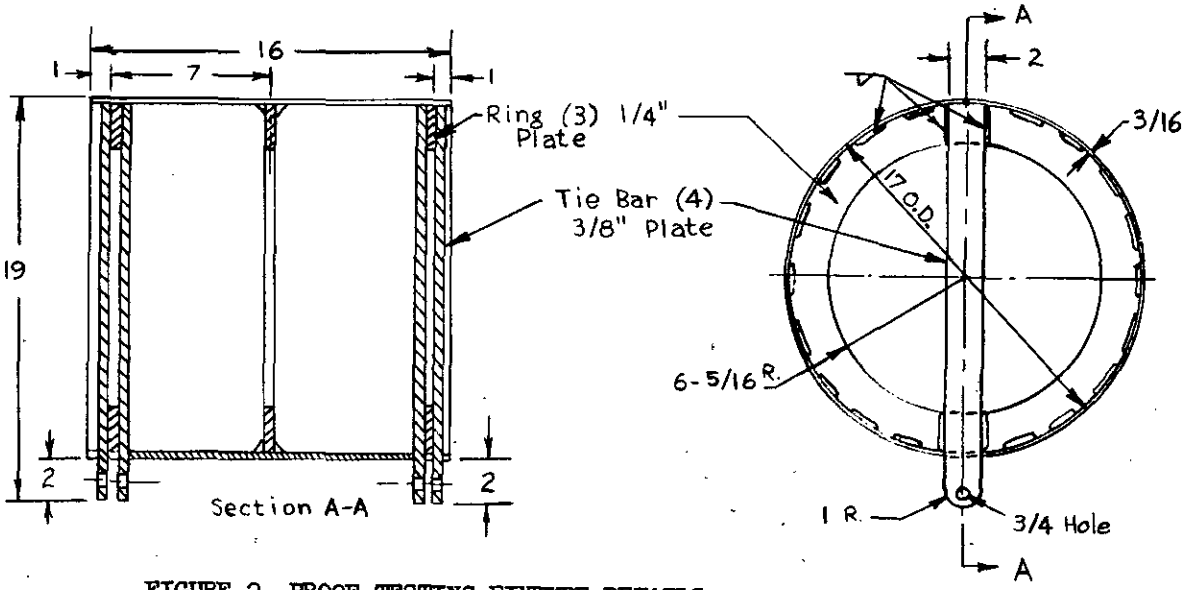
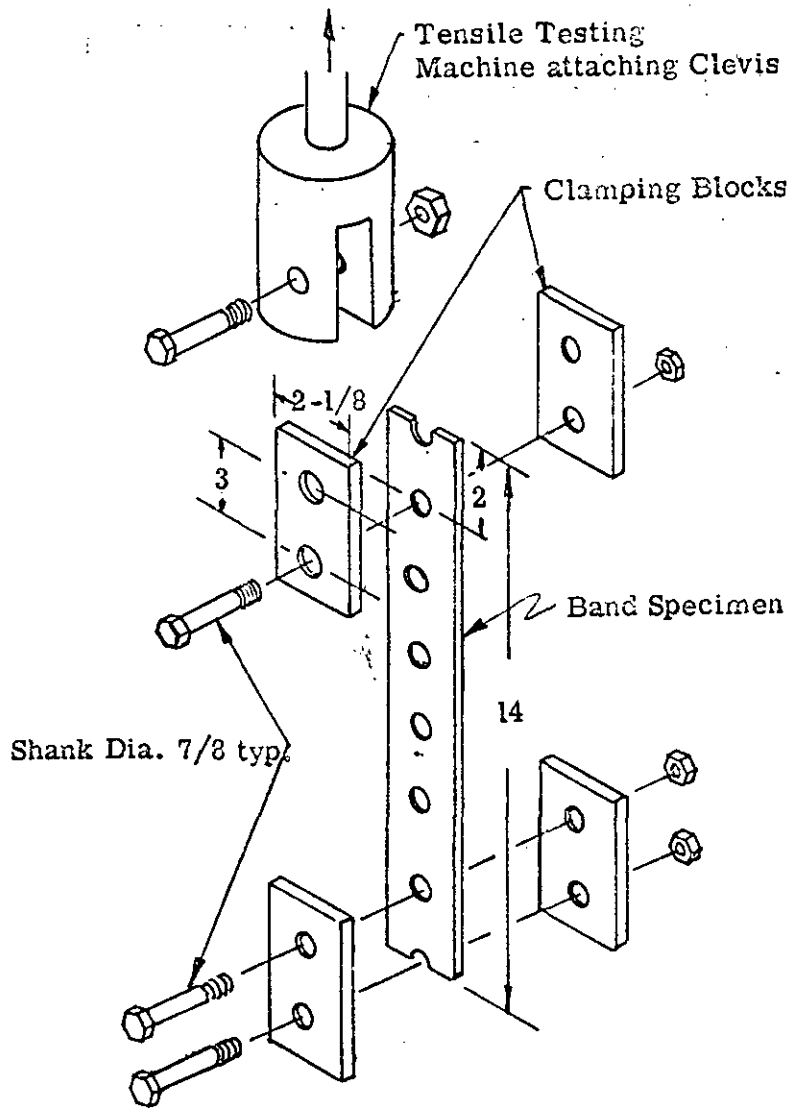
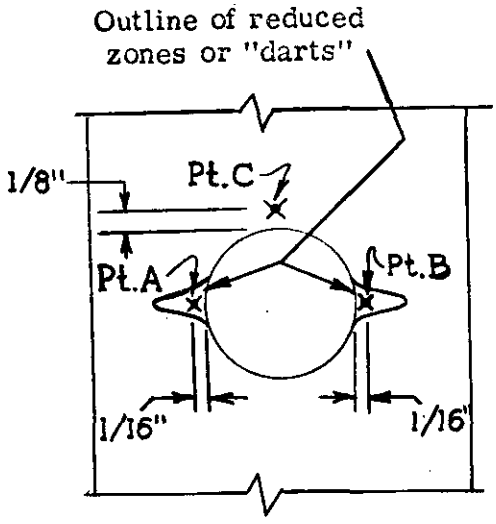


FIGURE 2. PROOF TESTING FIXTURE DETAILS

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Note: All bolts shall be tightened to 60 ft. lbs. torque



Note: Measurements of the reduce thickness shall be taken at points A and B, and the unaffected thickness at point C.

FIGURE 3. TEST SETUP FOR DETERMINING TENSILE STRENGTH & DUCTILITY

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